

Network Management is the process of monitoring, controlling, maintaining, and optimizing a computer network. It ensures that all network devices such as routers, switches, servers, and computers work properly and efficiently. Network management helps organizations maintain smooth communication, high performance, and strong security in their networks.

Objectives of Network Management

- To ensure **reliable network performance**
- To **detect and fix network problems quickly**
- To **maintain security and protect data**
- To **optimize network resources**
- To **monitor network usage and traffic**

Main Functions of Network Management (FCAPS Model)

1. Fault Management

Fault management deals with detecting, isolating, and correcting network problems.

- Identifies errors or failures in network devices.
- Generates alerts when problems occur.
- Helps administrators quickly fix issues to reduce downtime.

Example: Detecting a router failure or network cable disconnection.

2. Configuration Management

Configuration management handles the setup and maintenance of network devices.

- Keeps records of network configurations.
- Allows administrators to update or modify device settings.
- Ensures that devices are properly configured and compatible.

Example: Changing router settings or updating firmware.

3. Performance Management

Performance management monitors and improves the efficiency of the network.

- Measures network parameters such as bandwidth, traffic, response time, and throughput.
- Helps identify bottlenecks and improve network speed.

Example: Monitoring internet usage to prevent congestion.

4. Security Management

Security management protects the network from unauthorized access, attacks, and data breaches.

- Uses firewalls, authentication systems, and encryption.
- Controls user access to network resources.

Example: Blocking unauthorized users from accessing the network.

5. Accounting Management

Accounting management tracks network resource usage by users or departments.

- Maintains records of network usage.
- Helps in billing, cost control, and resource allocation.

Example: Monitoring how much bandwidth each user consumes.

Tools Used in Network Management

Some common tools and protocols used in network management include:

- **SNMP (Simple Network Management Protocol)**
- **Network monitoring software**
- **Log management systems**
- **Network analyzers**